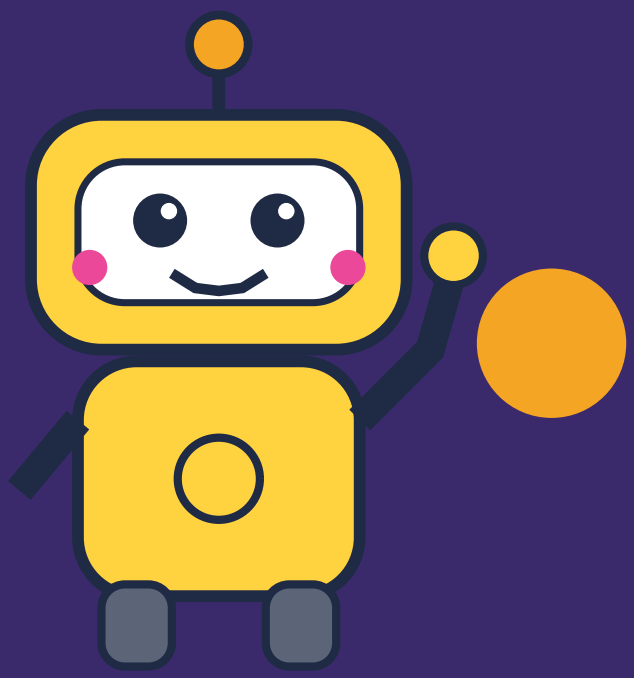
ASSESSMENT BOOK

Think Like a Coder

Chapter assessments for Class 4

14 one-page assessments, one for every chapter
10 marks each: choose, apply and think like a coder
Answers, marking guides and a class record sheet



For CBSE's Computational Thinking and AI curriculum, 2026-27

Goes with Think Like a Coder, Class 4 · Photocopiable for classroom use

SAMPLE EDITION · DRAFT

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Using these assessments

This book has one assessment for every chapter of *Think Like a Coder*, Class 4. Each assessment fits on one page, takes about 30 minutes and is marked out of 10. Use it after the chapter's last period, or as a revision check later in the term. All questions are new: they test the same skills as the workbook, not the same questions.

Section	What it checks	Questions	Marks
A	Knowing and understanding the chapter's ideas	3 multiple choice	3 x 1
B	Applying the idea to a short problem, with working	2 short answers	2 x 2
C	Thinking like a coder: write, trace, fix or explain a program or plan	1 task	1 x 3

Turning a score into a journal level

Score	Journal level	What it usually means
9-10	Extending	Solves new problems and explains the reasoning
7-8	Secure	Applies the idea correctly on their own
4-6	Developing	Understands with prompts; makes some slips
0-3	Beginning	Needs more practice with the chapter's idea

✓ Good practice

- ✓ Read the questions aloud for children who find reading hard: we are testing thinking, not reading.
- ✓ Give full marks for any correct method, even if it differs from the answer key; the marking guides list what to look for.
- ✓ Give part marks in Sections B and C for correct working, even when the final answer has a slip.
- ✓ Record the score on the class record sheet, and use it with your observations in the journal, not instead of them.
- ✓ Go over the test with the class: the most common mistakes are a lesson plan for the next period.

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Assessment 1 Shapes Around Us

Coder skill: Faces, edges and nets · Level B1 review · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 How many faces does a cube have? [1 mark]

a) 4 **b)** 6 **c)** 8 **d)** 12

2 The line where two faces meet is called ... [1 mark]

a) an edge **b)** a net **c)** a face

3 A matchbox is a cuboid. How many corners does it have? [1 mark]

a) 6 **b)** 8 **c)** 10 **d)** 12

Section B Answer in a few words or steps — 4 marks

4 A dice is a cube. Opposite faces add up to 7. Which number is opposite 2, and which is opposite 6? [2 marks]

5 The net of a closed cube has 6 squares. How many squares does the net of an open box with no lid have? Why? [2 marks]

Section C Think like a coder — 3 marks

6 Write exact steps to draw a rectangle 4 cm long and 2 cm wide with a ruler. [3 marks]

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Assessment 2 Hide and Seek

Coder skill: If-then-else choices · Level B3 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 In 'if it is raining then take an umbrella else wear a cap', what is the condition? [1 mark]

a) it is raining **b)** take an umbrella

c) wear a cap

2 Rule: if the number is more than 5 then say 'big' else say 'small'. What is said for 3? [1 mark]

a) big **b)** small **c)** nothing

3 An if-else inside a loop is checked ... [1 mark]

a) only once **b)** every time the loop repeats

c) never

Section B Answer in a few words or steps — 4 marks

4 Robo: repeat 4 times: move forward; if on a grey tile then say 'Splash' else say 'Step'. The tiles are white, grey, grey, white. What does Robo say? [2 marks]

5 Write an if-then-else rule for a street light. [2 marks]

Section C Think like a coder — 3 marks

6 For each number from 1 to 6: if the number is even then say 'clap' else say 'stamp'. Write what is said, and how many claps there are. [3 marks]

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Assessment 3 Patterns Around Us

Coder skill: Odd, even and checking for mistakes · Level B3 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 Which number is odd? [1 mark]

a) 14 **b)** 20 **c)** 27 **d)** 32

2 Odd + odd is always ... [1 mark]

a) odd **b)** even **c)** zero

3 A parity check makes each row have ... number of black coins. [1 mark]

a) an even **b)** an odd **c)** a big

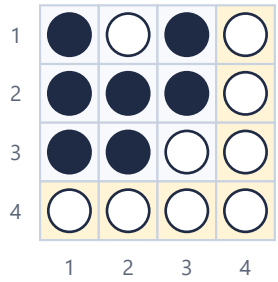
Section B Answer in a few words or steps — 4 marks

4 A row of coins is black, white, black, black, white. Add one check coin so the row has an even number of black coins. Black or white? Why? [2 marks]

5 Each row should have an even number of 1s. Row P: 1 1 0 0. Row Q: 1 0 1 1. Row R: 0 1 0 1. Which row has a mistake? [2 marks]

Section C Think like a coder — 3 marks

6 Every row and column had an even number of black coins. Then one coin was flipped. Which coin was it? Explain how you found it. [3 marks]



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Assessment 4 Thousands Around Us

Coder skill: Big numbers and sorting machines · Level B4 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 In 5,308, what does the 3 stand for? [1 mark]

a) 3 ones **b)** 3 tens **c)** 3 hundreds **d)** 3 thousands

2 Which number is the biggest? [1 mark]

a) 4,099 **b)** 4,909 **c)** 4,990 **d)** 4,900

3 A comparator gets 7 and 3. Which number goes to the upper wire? [1 mark]

a) 7 **b)** 3 **c)** both

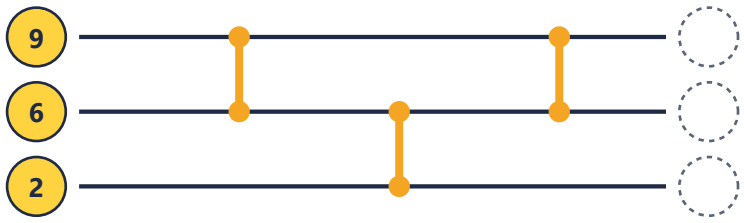
Section B Answer in a few words or steps — 4 marks

4 Write the number: 2 thousands, 0 hundreds, 5 tens and 9 ones. Then write it in words. [2 marks]

5 Order from smallest to biggest: 3,210; 3,120; 3,201. [2 marks]

Section C Think like a coder — 3 marks

6 Send 9, 6 and 2 through this network (top two wires, then bottom two, then top two). Write the order after each comparator. [3 marks]



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Assessment 5 Sharing and Measuring

Coder skill: Fractions and checking a condition · Level B3 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 A roti is cut into 4 equal parts. Each part is ... [1 mark]

a) one half **b)** one quarter **c)** one third

2 Two quarters make ... [1 mark]

a) one whole **b)** one half **c)** one third

3 Which is bigger? [1 mark]

a) 1/2 **b)** 1/4 **c)** they are equal

Section B Answer in a few words or steps — 4 marks

4 What is one quarter of 20 laddoos? What is three quarters of 20? [2 marks]

5 Rule: if all the parts are equal then say 'fair' else say 'not fair'. A cake is cut into 2 big and 2 small parts. What does the rule say? Why? [2 marks]

Section C Think like a coder — 3 marks

6 Mina has 16 marbles. She gives half to Ravi and a quarter to Asha. How many does Ravi get, how many does Asha get, and how many are left for Mina? [3 marks]

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Assessment 6 Measuring Length

Coder skill: Shortest paths and loops with turns · Level B4 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 1 metre = ... centimetres [1 mark]

a) 10 **b)** 100 **c)** 1,000

2 3 m 20 cm = ... cm [1 mark]

a) 32 **b)** 320 **c)** 3,020

3 'Repeat 4 times: forward 3, turn right' draws a ... [1 mark]

a) triangle **b)** square **c)** straight line

Section B Answer in a few words or steps — 4 marks

4 Robo is facing up. It turns right, then turns right again. Which way is it facing now? [2 marks]

5 A rope is 2 m long. You cut off 75 cm. How long is the rope now? [2 marks]

Section C Think like a coder — 3 marks

6 Write a loop with turns that draws a rectangle 4 steps long and 2 steps wide. [3 marks]

How did I do? Colour a face.

I can finish this test.	I can show my thinking.	I can check my answers.
		

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Assessment 7 The Cleanest Village

Coder skill: Tracing a machine step by step · Level B3 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 A lamp is ON. You press its toggle button once. Now it is ... [1 mark]

a) ON **b)** OFF **c)** broken

2 A toggle lamp starts OFF. You press it 3 times. Now it is ... [1 mark]

a) ON **b)** OFF

3 Writing down the state after every press is called ... [1 mark]

a) tracing **b)** guessing **c)** sorting

Section B Answer in a few words or steps — 4 marks

4 Lamps A and B start OFF. Button 1 flips lamp A. Button 2 flips both lamps. Press 2, 2, 1. Which lamps are ON at the end? [2 marks]

5 A gate light starts red. Each press switches it between red and green. After 6 presses, is it red or green? Why? [2 marks]

Section C Think like a coder — 3 marks

6 Lamps A and B start OFF. Button 1 flips A, button 2 flips both, button 3 flips B. Make a trace table for the presses 2, 1, 3, 2. [3 marks]

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Assessment 8 Weigh It, Pour It

Coder skill: If-else decisions and measuring steps · Level B3 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 1 kilogram = ... grams [1 mark]

a) 10 **b)** 100 **c)** 1,000

2 On a balance, the left pan goes down. So ... [1 mark]

a) the left side is heavier **b)** the right side is heavier

c) they are equal

3 The best unit to measure the water in a bucket is ... [1 mark]

a) litres **b)** kilograms **c)** metres

Section B Answer in a few words or steps — 4 marks

4 Rule: if bag A is heavier than bag B then put A on the shelf, else put B on the shelf. Bag A is 2 kg and bag B is 1,500 g. Which bag goes on the shelf? [2 marks]

5 You have a 4 L jug and a 1 L jug. How can you measure exactly 3 litres? [2 marks]

Section C Think like a coder — 3 marks

6 You have a 3 L jug and a 5 L jug. Write the steps to get exactly 1 litre. [3 marks]

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Assessment 9 Equal Groups

Coder skill: Loops inside loops · Level B4 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 A nested loop is ... [1 mark]

a) a loop inside a loop
b) a loop that never stops
c) two loops one after the other

2 Outer loop: repeat 2 times. Inner loop: repeat 5 times: draw a star. How many stars? [1 mark]

a) 7 **b)** 10 **c)** 25

3 An array has 4 rows of 6 chairs. How many chairs? [1 mark]

a) 10 **b)** 20 **c)** 24 **d)** 46

Section B Answer in a few words or steps — 4 marks

4 Write a nested loop that draws 3 rows of 5 dots. [2 marks]

5 18 chairs stand in an array, 6 in each row. How many rows are there? Write it as a multiplication. [2 marks]

Section C Think like a coder — 3 marks

6 Program: repeat 3 times: clap once, then (repeat 4 times: stamp). How many claps and how many stamps? What would you change to get 15 stamps? [3 marks]

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Assessment 10 Elephants, Tigers and Leopards

Coder skill: Clue tables and data · Level B3 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 In a clue table, a cross means ... [1 mark]

a) not possible **b)** true **c)** not sure

2 A row has crosses in every box but one. That box must be ... [1 mark]

a) true **b)** false **c)** empty

3 A park has 12 elephants, 7 tigers and 9 leopards. Which animal has the fewest? [1 mark]

a) elephants **b)** tigers **c)** leopards

Section B Answer in a few words or steps — 4 marks

4 Use the same park. How many more elephants than leopards are there, and how many animals in all? [2 marks]

5 Ravi and Meena own a cat and a dog, one pet each. Ravi does not own the cat. Who owns which pet? [2 marks]

Section C Think like a coder — 3 marks

6 Asha, Bilal and Chitra each like a different fruit: apple, banana or mango. Asha likes neither apple nor mango. Bilal does not like apple. Who likes which fruit? [3 marks]

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Assessment 11 Fun with Symmetry

Coder skill: Mirror patterns and repeat with turns · Level B4 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 How many lines of symmetry does a square have? [1 mark]

a) 1 **b)** 2 **c)** 4

2 Which capital letter has a line of symmetry going straight down? [1 mark]

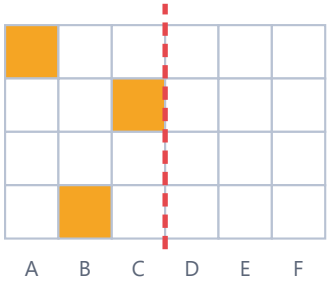
a) A **b)** F **c)** G **d)** R

3 A mirror image is ... [1 mark]

a) a flipped copy **b)** a bigger copy **c)** a moved copy

Section B Answer in a few words or steps — 4 marks

4 The red line is the mirror. A square in column A has its partner in column ___. A square in column C has its partner in column ___. [2 marks]



5 Which word has a line of symmetry straight down its middle: MOM or DAD? Why? [2 marks]

Section C Think like a coder — 3 marks

6 Robo draws one petal, then turns right. Write a loop that draws a flower with 4 petals all the way round. Then draw the flower. [3 marks]

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Assessment 12 Ticking Clocks and Turning Calendar

Coder skill: Repeat until and cycles · Level B4 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 1 hour = ... minutes [1 mark]

a) 30 **b)** 60 **c)** 100

2 Today is Friday. What day will it be after 7 days? [1 mark]

a) Friday **b)** Saturday **c)** Thursday

3 A 'repeat until' loop stops when ... [1 mark]

a) the condition becomes true **b)** it has run 10 times

c) it gets tired

Section B Answer in a few words or steps — 4 marks

4 Start at 2:00. Repeat until the clock shows 3:30: add 15 minutes. How many times does the loop run? [2 marks]

5 Today is Tuesday. What day will it be after 10 days? [2 marks]

Section C Think like a coder — 3 marks

6 Write a repeat-until loop to fill a 10 litre bucket with a 2 litre mug. How many times does it run? [3 marks]

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Assessment 13 The Transport Museum

Coder skill: Decision trees: sorting by questions · Level B3 · 30 minutes · 10 marks

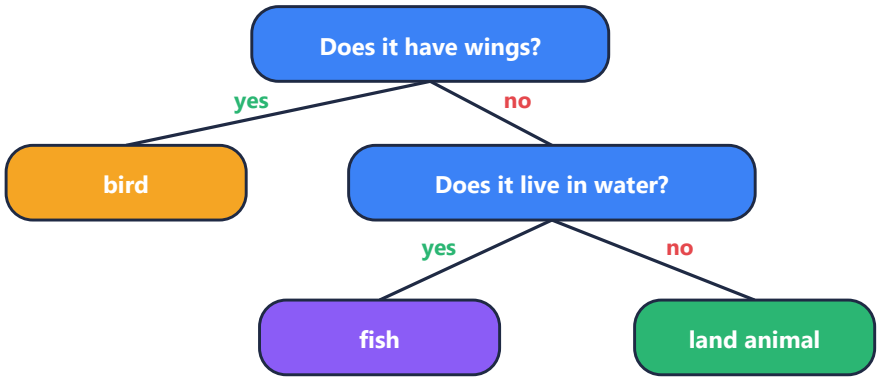
Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 A decision tree asks questions that have ... answers. [1 mark]

a) yes or no **b)** any number of **c)** colour

2 In this tree, an animal with no wings that lives in water is a ... [1 mark]



a) bird **b)** fish **c)** land animal

3 To classify means to ... [1 mark]

a) put things into the right group **b)** count very fast

c) draw a picture

Section B Answer in a few words or steps — 4 marks

4 Use the tree above. Where does a cow end up? Where does a crow end up? [2 marks]

5 Write one yes/no question that splits pencil, apple, ball and banana into two groups of 2. [2 marks]

Section C Think like a coder — 3 marks

6 Make a decision tree with 2 questions that sorts a car, a bicycle and a boat. [3 marks]

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Assessment 14 Data Handling

Coder skill: Collecting and showing data · Level B2 review · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 Four lines with one line across them (tally marks) mean ... [1 mark]

a) 4 **b)** 5 **c)** 6

2 In a bar graph, the tallest bar shows ... [1 mark]

a) the most popular **b)** the least popular

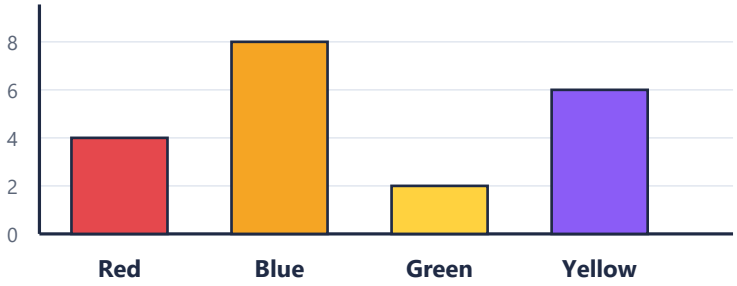
c) the first item

3 Asking many people the same question is called a ... [1 mark]

a) survey **b)** story **c)** sum

Section B Answer in a few words or steps — 4 marks

4 Which colour is the most popular? How many more children chose Blue than Green? [2 marks]



5 Use the same graph. How many children answered in all? [2 marks]

Section C Think like a coder — 3 marks

6 Draw tally marks for: cricket 7, football 4, kho-kho 9. Then write one question this data can answer, and its answer. [3 marks]

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Answers and marking guide

Accept any correct method. Part marks are shown in brackets for Sections B and C.

1. Shapes Around Us

1. b) 6
2. a) an edge
3. b) 8
4. 5 is opposite 2; 1 is opposite 6 (1 mark for each)
5. 5, because the lid is missing (1 mark for 5, 1 mark for the reason)
6. 1. Draw a straight line 4 cm long. 2. At each end draw a line 2 cm long, straight up. 3. Join the two tops with a line 4 cm long. (1 mark for each clear step that gives a length or position)

2. Hide and Seek

1. a) it is raining
2. b) small
3. b) every time the loop repeats
4. Step, Splash, Splash, Step (2 marks; 1 mark if three words are right)
5. If it is dark then switch the light on, else switch it off (1 mark for a sensible condition, 1 mark for both actions)
6. stamp, clap, stamp, clap, stamp, clap; 3 claps (2 marks for the list (1 mark with one slip), 1 mark for 3 claps)

3. Patterns Around Us

1. c) 27
2. b) even
3. a) an even
4. Black, because there are 3 black coins and 3 is odd (1 mark for black, 1 mark for the reason)
5. Row Q (it has three 1s) (2 marks; 1 mark for counting the 1s correctly in at least two rows)
6. Row 2, column 1. Row 2 has 3 black coins and column 1 has 3 black coins; they cross at that coin. (1 mark for the odd row, 1 mark for the odd column, 1 mark for the coin where they cross)

4. Thousands Around Us

1. c) 3 hundreds
2. c) 4,990
3. b) 3
4. 2,059; two thousand fifty-nine (1 mark for each)
5. 3,120; 3,201; 3,210 (2 marks; 1 mark if one number is out of place)
6. After 1: 6, 9, 2. After 2: 6, 2, 9. After 3: 2, 6, 9. (1 mark for each comparator)

5. Sharing and Measuring

1. b) one quarter
2. b) one half
3. a) 1/2
4. 5 and 15 (1 mark for each)
5. Not fair, because the parts are not equal (1 mark for 'not fair', 1 mark for the reason)
6. Ravi 8, Asha 4, Mina 4 (1 mark for each)

6. Measuring Length

1. b) 100
2. b) 320
3. b) square
4. down (2 marks)
5. 125 cm (1 m 25 cm) (2 marks; 1 mark for changing 2 m to 200 cm)
6. Repeat 2 times: forward 4, turn right, forward 2, turn right. (1 mark for repeat 2 times, 1 mark for both side lengths, 1 mark for the turns)

7. The Cleanest Village

1. b) OFF
2. a) ON
3. a) tracing
4. Only lamp A is ON (A ON, B OFF) (2 marks; 1 mark for a correct trace with one slip)
5. Red, because 6 is even and every 2 presses bring it back (1 mark for red, 1 mark for the reason)
6. start OFF OFF; 2: ON ON; 1: OFF ON; 3: OFF OFF; 2: ON ON (3 marks for all four rows; 2 marks for three; 1 mark for two)

8. Weigh It, Pour It

1. c) 1,000
2. a) the left side is heavier
3. a) litres
4. Bag A (2 kg = 2,000 g, more than 1,500 g) (1 mark for changing to grams, 1 mark for bag A)
5. Fill the 4 L jug, then pour from it to fill the 1 L jug. 3 L are left. (2 marks; 1 mark for a sensible start)
6. Fill the 3 L jug and pour it into the 5 L jug. Fill the 3 L jug again and pour into the 5 L jug until it is full (2 L go in). 1 L is left in the 3 L jug. (3 marks for steps that work; 2 marks if one step is missing; 1 mark for a sensible start)

9. Equal Groups

1. a) a loop inside a loop
2. b) 10
3. c) 24
4. Repeat 3 times: (repeat 5 times: draw a dot), then start a new row (1 mark for the outer loop, 1 mark for the inner loop)
5. 3 rows; 3 x 6 = 18 (1 mark for each)
6. 3 claps and 12 stamps. Change the inner loop to repeat 5 times. (1 mark for claps, 1 mark for stamps, 1 mark for the change)

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10. Elephants, Tigers and Leopards

1. a) not possible
2. a) true
3. b) tigers
4. 3 more; 28 animals (1 mark for each)
5. Ravi owns the dog; Meena owns the cat (1 mark for each)
6. Asha: banana. Bilal: mango. Chitra: apple. (1 mark for each person)

11. Fun with Symmetry

1. c) 4
2. a) A
3. a) a flipped copy
4. F and D (1 mark for each)
5. MOM, because each half is a flipped copy of the other; D is not symmetric that way (1 mark for MOM, 1 mark for the reason)
6. Repeat 4 times: draw a petal, turn right. The drawing is a flower with 4 petals, one pointing each way. (1 mark for repeat 4 times, 1 mark for 'draw a petal, turn right' inside the loop, 1 mark for the drawing)

12. Ticking Clocks and Turning Calendar

1. b) 60
2. a) Friday
3. a) the condition becomes true
4. 6 times (2 marks; 1 mark for listing the times with one slip)
5. Friday (2 marks; 1 mark for seeing that 7 days brings back Tuesday)
6. Repeat until the bucket is full: pour in one mug (2 L). It runs 5 times. (1 mark for the loop, 1 mark for the condition, 1 mark for 5)

13. The Transport Museum

1. a) yes or no
2. b) fish
3. a) put things into the right group
4. cow: land animal; crow: bird (1 mark for each)
5. For example: Can you eat it? (1 mark for a yes/no question, 1 mark if it splits them 2 and 2)
6. Does it go on water? Yes: boat. No: Does it have an engine? Yes: car. No: bicycle. (1 mark for each good question, 1 mark if all three end in the right place)

14. Data Handling

1. b) 5
2. a) the most popular
3. a) survey
4. Blue; 6 more (1 mark for each)
5. 20 (2 marks; 1 mark for reading all the bars correctly)
6. Tallies: 5+2, 4, 5+4. For example: Which game is most popular? Kho-kho. (1 mark for the tallies, 1 mark for a question, 1 mark for its answer)

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Class record sheet

Class 4 ____ Section ____ Term ____. Write each score out of 10. You may photocopy this page.

Student	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Avg

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